in terms of the relative concentration of phase points. The density may be viewed as depending both on time and space. When discussing the changes in the density over time, we are taking a *macroscopic* perspective. The density itself can be viewed as another type of state, a *macroscopic state* or *macrostate*. In the case of Hamiltonian dynamics, the evolution of the density is governed by Liouville's equation. We discuss this equation below. The behavior of the evolving point set is slightly complicated by the fact that each trajectory is confined to a particular level set of energy. The corresponding points of two trajectories evolving on different level sets would be expected to differ for all time in a way that reflects the geometry of the energy function. To simplify the discussion we initially consider the dependencies within a particular energy level set only, however, at the end of the chapter we discuss the situation which commonly arises when using a symplectic integrator: the energy surface sampled by the discretization method is different from the desired energy surface.

Our treatment of statistical mechanics is abbreviated and specialized as our focus is ultimately on certain types of systems frequently arising in molecular dynamics and their numerical treatment. We refer the interested reader to more detailed treatments including [57, 70, 168, 197, 264, 297, 372].

5.1 The Solution Operator

In Chap. 3, we introduced the Lie derivative of a function g of the phase variables with respect to a dynamical system $\dot{z} = f(z)$:

$$\mathcal{L}_f g = f \cdot \nabla g.$$

This is obtained by differentiating a composite function $g(z(t))$ with respect to t, along a solution trajectory. That is, describing a trajectory in terms of the flow map by $z(t) = \mathscr{F}_t(\zeta)$, $t \in \mathbb{R}$, we have

$$\frac{\mathrm{d}}{\mathrm{d}t} g(\mathscr{F}_t(\zeta)) = (\mathcal{L}_f g)(\mathscr{F}_t(\zeta)).$$

Replacing ζ by z (it is an arbitrary point), define a function $\tilde{g}(z,t) \stackrel{\text{def}}{=} g(\mathscr{F}_t(z))$. The partial derivative of $\tilde{g}(z,t)$ with respect to t is

$$\frac{\partial \tilde{g}}{\partial t}(t,z) = \nabla g(\mathscr{F}_t(z)) \cdot \frac{\mathrm{d}}{\mathrm{d}t} \mathscr{F}_t(z) = \nabla g(\mathscr{F}_t(z)) \cdot f(\mathscr{F}_t(z)) = (\mathcal{L}_f g)(\mathscr{F}_t(z)). \tag{5.1}$$

On the other hand, viewing $\mathcal{L}_f$ as an operator on the phase variables only, we have

$$\begin{aligned}(\mathcal{L}_f \tilde{g})(z) &= \boldsymbol{f}(z) \cdot \nabla_z \left[g(\mathscr{F}_t(z))\right] \\ &= \boldsymbol{f}(z) \cdot \left[\mathscr{F}_t'(z)^T \nabla g(\mathscr{F}_t(z))\right] \\ &= \left[\mathscr{F}_t'(z)\boldsymbol{f}(z)\right] \cdot \nabla g(\mathscr{F}_t(z)) \\ &= \boldsymbol{f}(\mathscr{F}_t(z)) \cdot \nabla g(\mathscr{F}_t(z)) \\ &= (\mathcal{L}_f g)(\mathscr{F}_t(z)).\end{aligned}$$

This is the same as the right hand side of (5.1). This means that $\tilde{g}$ satisfies the equation

$$\frac{\partial \tilde{g}}{\partial t} = \mathcal{L}_f \tilde{g}. \tag{5.2}$$

This is a time-dependent, linear (hyperbolic) partial differential equation. The steady state solutions of this equation (defined by $\partial \tilde{g}/\partial t = 0$) are of the form $\tilde{g}(z,t) = \varphi(z)$ where φ satisfies $\mathcal{L}_f \varphi(z) \equiv 0$. These are just the first integrals of the ordinary differential equation system $\dot{z} = \boldsymbol{f}(z)$.

It is intuitively appealing to define the solution of (5.2) by

$$\tilde{g}(z,t) = G_t \tilde{g}(z,0), \qquad \text{where} \quad G_t = e^{t\mathcal{L}_f}.$$

Let us refer to G_t as the *solution operator* associated to the differential equation $\dot{z} = \boldsymbol{f}(z)$. Calculating the exponential operator $e^{t\mathcal{L}_f}$ can be thought of as tantamount to solving a linear partial differential equation for given (Cauchy) initial data.

The projection onto the ith coordinate, denoted ∂_i, is itself a scalar valued function of z ($\partial_i(z) = z_i$). Calculating directly we have $\mathcal{L}_f z_i = f_i$. The interpretation of the solution of the initial value problem in terms of the solution operator G_t is given by

$$\left(e^{t\mathcal{L}_f} \partial_i\right)(z(0)) = z_i(t).$$

The notion of the solution operator and the concept of flow map are essentially equivalent descriptions of the dynamics of the system.

In physics treatments, the Liouvillian is often denoted $i\mathcal{L}$. This may seem natural since (a) it allows a formal correspondence with the Schrödinger equation and between the propagators of quantum mechanics and classical mechanics, and (b) for Hamiltonian dynamics, as we shall see, the Liouvillian is skew-adjoint (in a certain sense skew symmetric) and the inclusion of i explicitly calls attention to this fact. However, we feel it is more natural to omit the i in a treatment that includes study of both stochastic and deterministic models.